

All-Around Defender: What It Really Takes to Win in Cybersecurity Today

Based on Security 530: Defensible Security Architecture and Engineering:
Implementing Zero Trust for the Hybrid Enterprise

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Ismael Valenzuela

- SEC530 Author and Senior instructor
- +14 yrs teaching for SANS DFIR & Cyber Defense curriculum internationally
- GSE # 132, plus 11 other GIAC certifications
- VP Labs, Threat Research and Intelligence at Arctic Wolf
- Former Foundstone, Intel/McAfee, BlackBerry/Cylance
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Once upon a time...

Miercoles 20 de Agosto de 2003

Inicio / Acceso Client



Seguridad en Redes e Internet

Network and Internet Security



Empresa

Servicios Profesionales

Productos y soluciones

Alianzas

Noticias

En Prensa Especializada
Administraciones Públicas
Legislación
Banca Electrónica
Educación
Industria

El ataque de los SuperGusanos.

Todos los expertos en seguridad coinciden, la amenaza que los ordenadores se está multiplicando tanto en sofisticación como en volumen.

Bajo el título **"Test de Intrusión y Hacking Ético: ¿por qué necesita uno?"** explica en qué consiste un Test de Intrusión y qué se conoce por Hacking Ético. Una vez explicados estos conceptos, reflexiona sobre las consecuencias de un fallo en la seguridad de los sistemas de una compañía y las ventajas que supone el contar con una plataforma informática correctamente auditada y securizada.

Si desea leer dicho artículo, puede descargarse las páginas en formato JPG desde los siguientes enlaces:

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web

Nuevas tecnologías

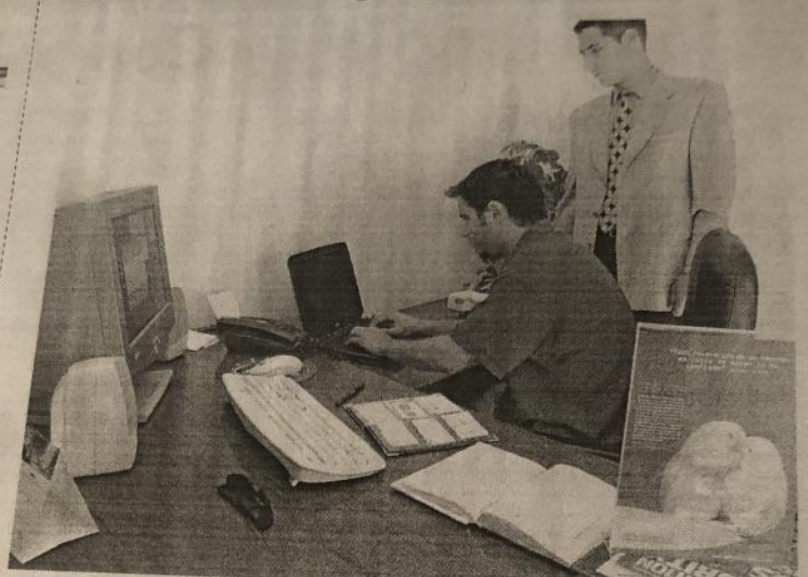


Imagen de las instalaciones de la empresa G2 Security en la capital
FERNANDO GONZÁLEZ

G2 Security amarra la red de las empresas

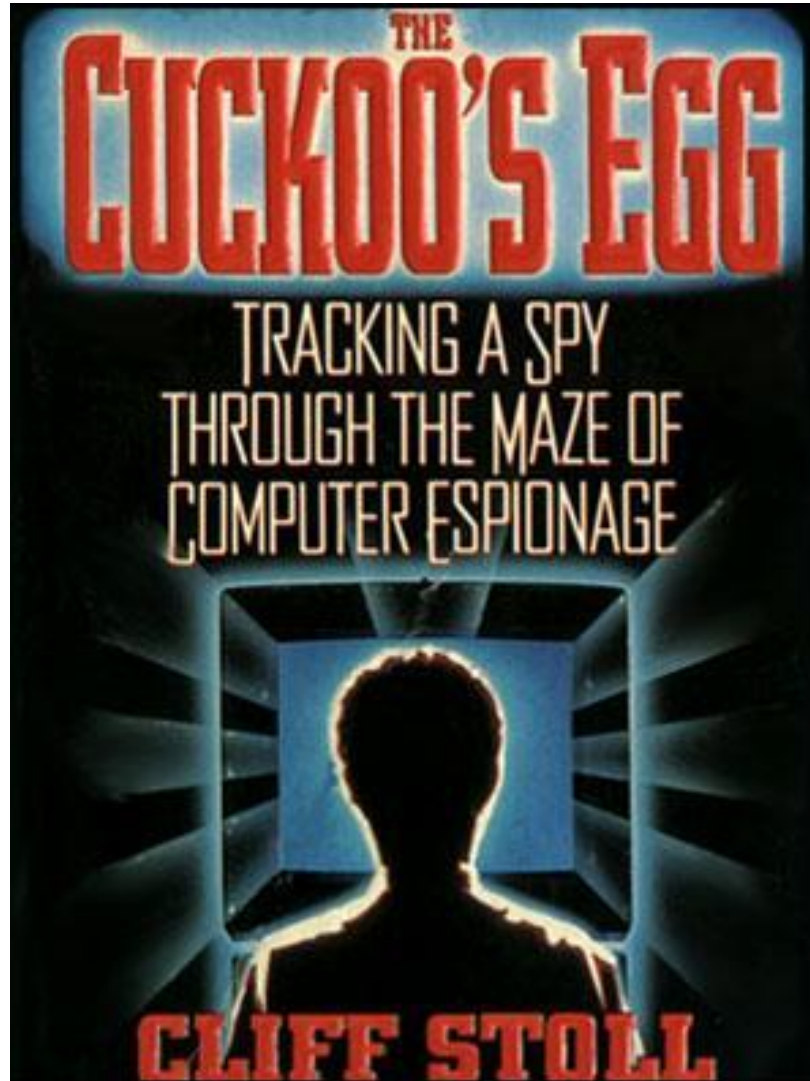
La firma de seguridad informática multiplica su actividad ante la mayor preocupación de los empresarios por la confidencialidad de su negocio

A. J. LÓPEZ

«Una persona puede acceder a la información confidencial de una empresa con la misma facilidad que puede cruzar la puerta de entrada de esa compañía», afirma con rotundidad el director general de G2 Security, Ismael Valenzuela. Este joven directivo repite varias veces que su discurso no pretende ser alarmista, aunque reconoce que las acciones fue uno de los motivos que llevó al grupo G2 –al que también pertenecen G2 Producciones, Topfutbol.com y G2 Network– a crear esta sociedad, que en apenas un año de funcionamiento se ha hecho un nombre.

La compañía negocia un contrato con el gobierno para mantener con una «importante operadora nacional» para preservar la seguridad de sus operaciones, así como de los servicios que presta. Por el momento, Valenzuela prefiere no dar más detalles sobre el próximo cliente de esta firma asentada en la capital, aunque adelanta que se trata de una de las empresas más importantes del país.

39 years of Cyber Defense



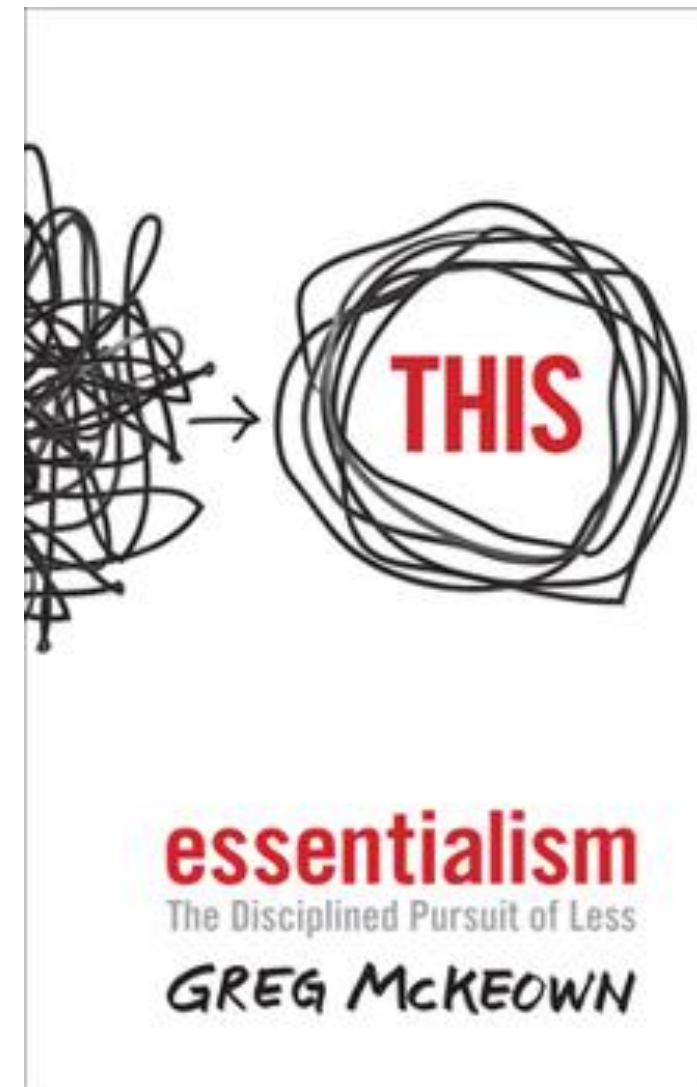
Why are we taken by surprise every time?

- Addressing threats based on products, not outcomes (aka. “shiny object syndrome”)
- Too much prevention and compliance focus
- Security is treated as “digital”; all or nothing
- Lack of **actionable** knowledge of the adversary
- You have to do more with less (again)



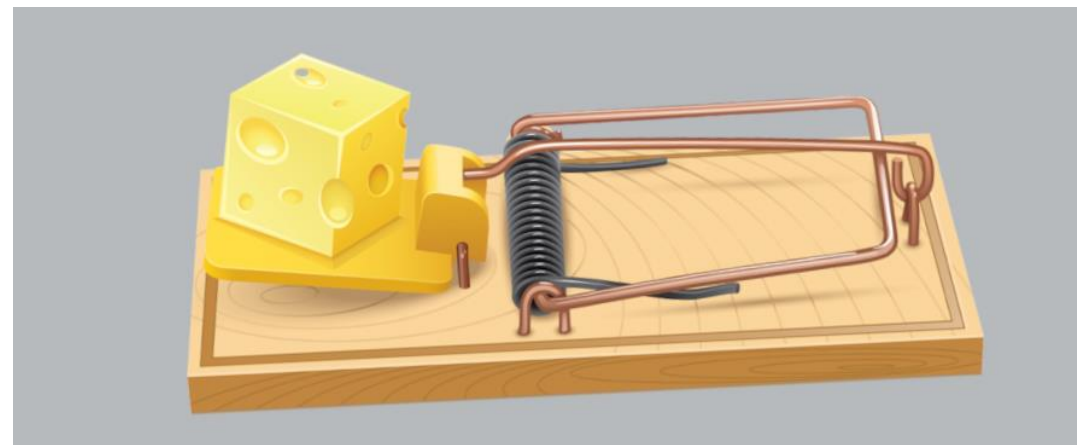
Bad news: the 'less' part is non-negotiable

- But, do we need more doing?
- What we want is more **impact**, more bang for the buck.
- Perhaps we can achieve that by doing **less**
- The discipline at work here is **prioritization**.
- The word ***priority*** entered English in the 1400s. It was singular, meaning “the very first thing” or “the thing that comes before.”
- After **1900**: we created *priorities* → **deluding ourselves into thinking we could have multiple “first” things; many #1s**
- Result: useful in theory, but also confusing, since if everything is a priority, nothing truly is.



Sell the outcome, not the product

- Focus on the ‘selling the problem you will solve’, the product or the tool you think you need.
- In some cases, **you may need to create a need.**
- Example:
 - You need resources to re-architect your network in a **Zero Trust** like manner
 - **Focus on building visibility** for detection and investigation
 - Pick a strategic use case, and use Security Onion on legacy HW



Be 'obsessed' on communicating VALUE

- ***Ironically, doing security right can put your job at risk.***
- Good security is boring: invisible, no incidents, no drama.
- Bad security is loud: breaches, late-night heroics, “superheroes” saving the day.
- Executives don't see invisibility; they see cost.
- When you don't obsessively communicate value, your budget may be the first to get cut.



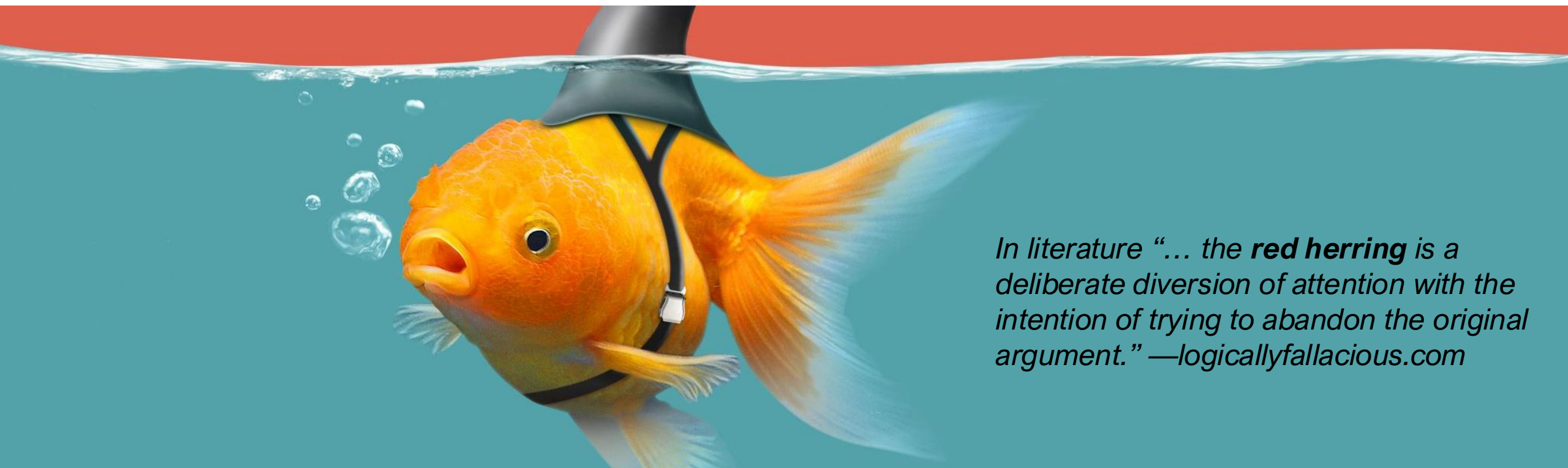
**Boring is
good
Boring
means
your safe.**

AI Hype Cycle Survival Guide

- I've seen hype cycles before. In 2001, PKI was going to solve world hunger!
- Watch for these pitfalls:
 - POCs that look great but never scale
 - IBM Watson: In 2011, IBM Watson was going to revolutionize healthcare and finance after it won Jeopardy
 - In AI (and security, too), the hardest part is not the **prototype**, it's the **last mile**
 - Dollars shifted away from proven defenses, new attack surfaces created
- Where AI (LLMs) helps:
 - Threat intel collection
 - Detection engineering
 - Rule creation
 - Integrations and parsers
 - Data reduction pipelines, etc.
 - Taking the adversary by surprise!



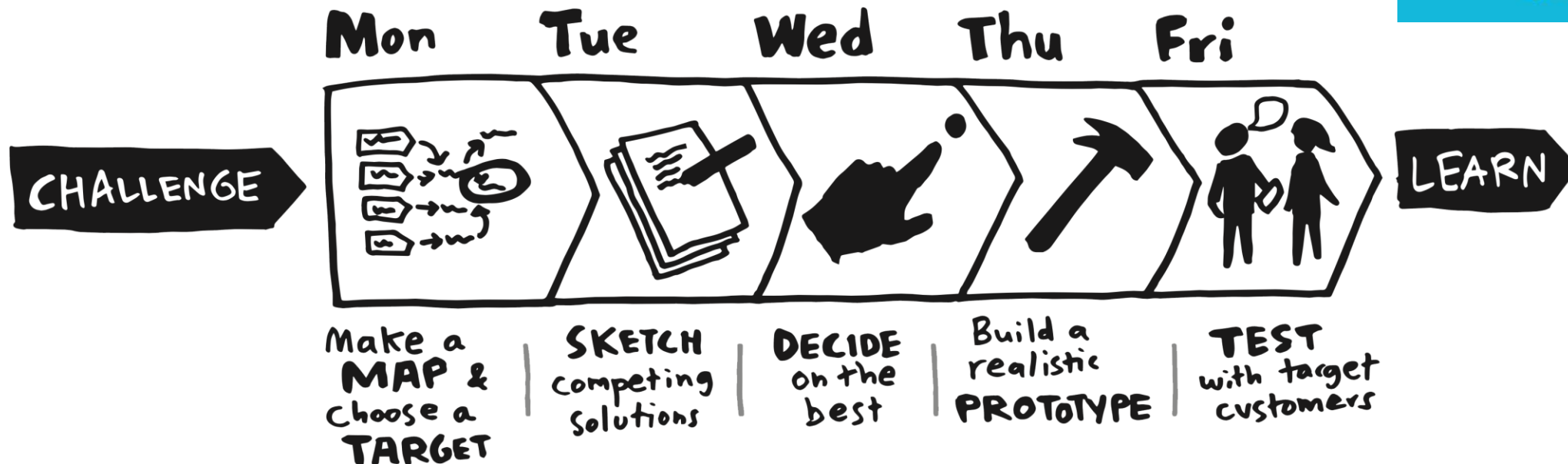
Think red, act blue: taking the adversary by surprise



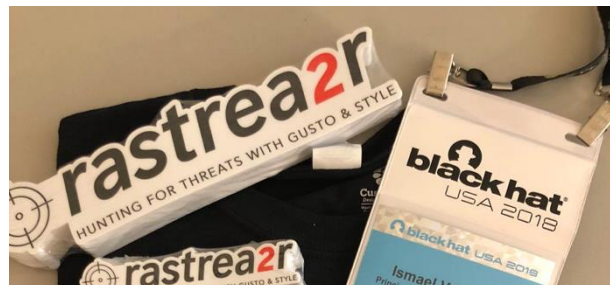
*In literature “... the **red herring** is a deliberate diversion of attention with the intention of trying to abandon the original argument.” —logicallyfallacious.com*

Plan & Execute Fast – Lessons from Google's Sprint

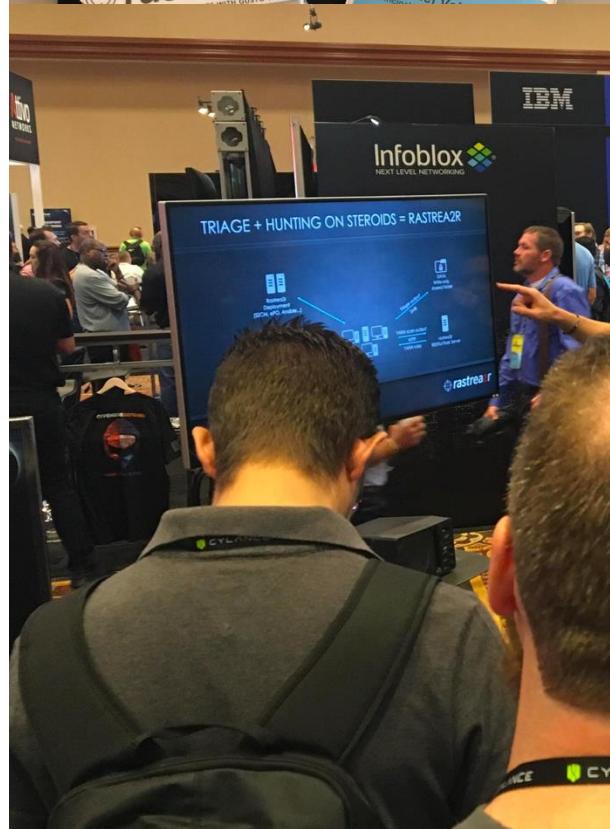
- Don't overplan: test ideas quickly to learn what works.
- Why it matters in security:
 - Speed reduces risk of wasted effort.
 - Small, fast experiments → actionable outcomes.
 - Better to learn fast than to polish the wrong solution.



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steroids = rastrea2r };
communication flows



black hat
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Since he founded one of the first IT Security consultancies in Spain, Ismael Valenzuela has participated as a security professional in numerous projects across the globe over the past 18 years. As a top cybersecurity expert with strong technical background and deep knowledge of penetration testing, security architectures, intrusion detection and computer forensics, Ismael has provided security consultancy, advice and guidance to large government and private organizations, including major EU Institutions and US Government Agencies. Prior to his current role as Principal Engineer at McAfee, where he leads research on threat hunting using machine-learning and expert-system driven investigations, Ismael led the delivery of SOC, IR & Forensics services for the Foundstone Services team within Intel globally. Previously, Ismael worked as Global IT Security Manager for ISOFT Group Ltd, one of the world's largest providers of healthcare IT solutions, managing their security operations in more than 40 countries. He holds a bachelor's degree in computer science from the University of Malaga (Spain), is certified in business administration, and holds many professional certifications. These include the highly regarded GIAC Security Expert (GSE #132) in addition to GREM, GCFA, GCIA, GCIH, GPCN, GCUX, GCWN, GWAPT, GSNA, GMON, CISSP, ITIL, CISM, and IRCA 27001 Lead Auditor from Bureau Veritas UK.

SESSIONS WITH THIS SPEAKER

- SPONSORED SESSION SPEAKER** | *Intelligence Driven Defense: Successfully Embedding Cyber Threat Intel in Security Operations* - Aug 08, 16:10
- ARSENAL** | *rastrea2r (reloaded!): Collecting & Hunting for IOCs with Gusto and Style* - Aug 09, 10:00

DIGITAL FORENSICS | INCIDENT RESPONSE

Speak the business vendor language

- Vendors don't sell configs: they sell magic
- Example:
 - ❌ *"We'll roll out MFA, segmentation, federation, reverse proxies, logging, and least privilege."* (a cure for insomnia)
 - ✅ *"We're implementing Conditional Access and a Zero Trust Architecture."* (now that sounds board-worthy)
- It's the same thing, but one gets you budget, the other gets you ignored.
- Package your work as strategy, not plumbing.



Upgrade Your Firmware

- Know your weaknesses, and patch them!
- Invest in training, mentorship, and growth to close gaps.
- Don't forget the human side: build your human API: connect, share, lean on peers ("phone-a-friend" networks)
- And remember: you are more important than your job
 - protect your health, delegate, disconnect, play music, spend time with family and friends.



“An All-Around Defender isn’t someone who knows everything. It’s someone who makes the most of what they have, prioritizes ruthlessly, and builds a mindset for the long run”

Fireside Chat



Associate Instructor

Principal Subject Matter
Expert & Research Lead at
Maltego Technologies



Senior Instructor

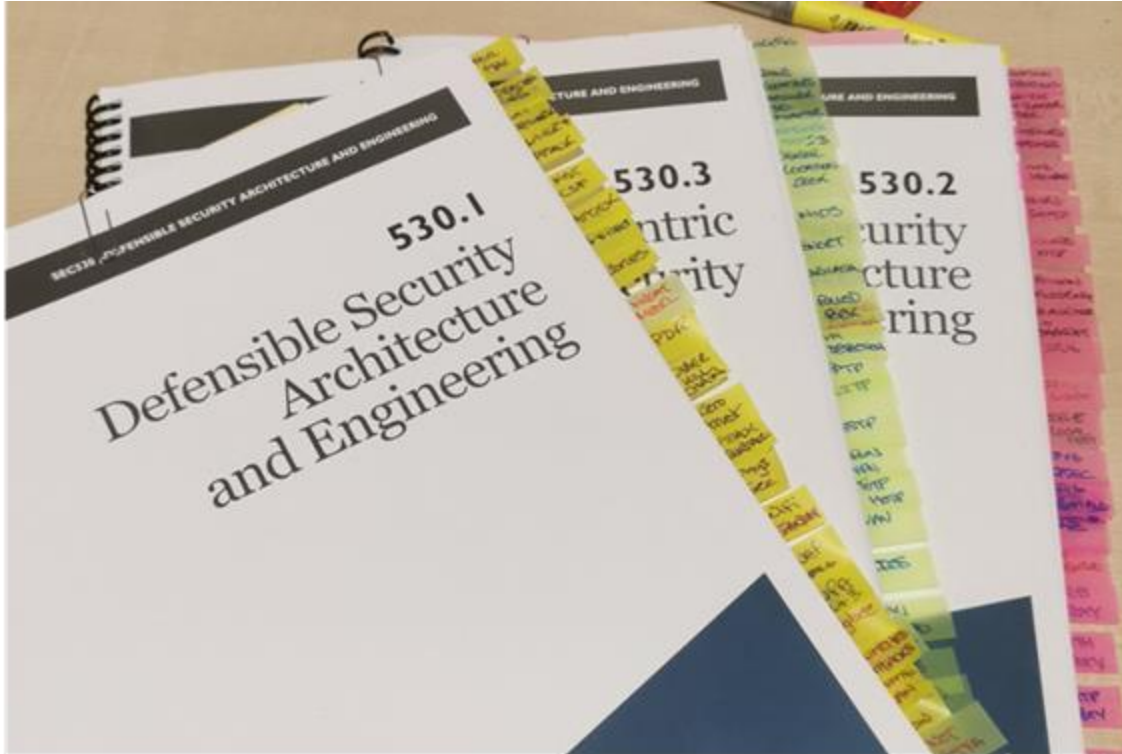
Founder at One eSecurity



Senior Instructor

Vice President Labs,
Threat Research &
Intelligence at Arctic Wolf

SEC530: Defensible Security Architecture and Engineering



“I would highly recommend this for any business and organizations [...] to fully understand why this attitude of Zero Trust needs to be taken into consideration. This covers areas that CISSP or Sec+ would not.” – SEC530 student, DHS-CISA

“The labs are absolutely rock solid. I've been around the industry a while and **never experienced labs this good.** Highly commendable.” – Michael Curran, Austrade

***#SEC530** is absolutely recommended for anyone who is working as Cyber Security Engineer / Analyst / Architect and wants to become an **#AllRoundDefender**. Great content and learning towards the journey to **Zero Trust Architecture** – Arindham Laha, Lead Architect, The TJX Companies*

<https://www.sans.org/cyber-security-courses/defensible-security-architecture-and-engineering/>